

Validating RADIS against the Korkin et al. (2025) line-by-line absorption codes (gcell & aspect)

Korkin, Sayer, Ibrahim & Lyapustin, JQSRT 337, 109345 (2025)

"A practical guide to coding line-by-line trace gas absorption in Earth's atmosphere"

<https://doi.org/10.1016/j.jqsrt.2025.109345> | Reference codes: github.com/korkins/aspect_gcell

WHAT I DID

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Goal: reproduce the results of the paper's open-source C codes using RADIS (a Python line-by-line radiative-transfer code) and overplot both on the same graph to check the match. The paper ships two codes and their reference output (benchmark) files:

- * gcell - absorption in a homogeneous gas cell
- * aspect - absorption in Earth's atmosphere, with thermodynamic profiles varying with height

Steps taken

1. Cloned the authors' repository (github.com/korkins/aspect_gcell) to obtain BOTH the source codes and their committed benchmark outputs (the actual C-code results, used here as ground truth - not a third-party library).
2. gcell reproduction. Ran RADIS with the exact cell conditions from the paper's input (*.inp) files (HITRAN database, Voigt line shape - same physics as the paper), for the two benchmarks:
 - O2 A-band : 13006-13166 cm⁻¹, T=296 K, p=0.7145 atm, L=1633.6 cm, pure O2
 - CH4 2.3um : 4081.901-4505.699 cm⁻¹, T=296 K, p=1.0 atm, L=8 cm, pure CH4Overplotted RADIS optical thickness tau(nu) directly on the paper's gcell output.
3. aspect reproduction using the RADIS STACK function. The atmospheric code integrates the absorption optical depth from the top of the atmosphere (TOA) down to an altitude z, using the US Standard 1976 profile (T, p, density vs height) taken from the paper's src/hprofiles.h. I rebuilt this as a stack of homogeneous slabs - one slab per profile level-gap - and combined them with RADIS's SerialSlabs (the 'stack' operator). Stacking slabs multiplies the transmittances, which is identical to ADDING optical depth, so the partial column tau(nu,z) from TOA to z is the sum of all slab optical depths above z. Verified numerically:
 - max | SerialSlabs(slabs) - sum(slab optical depths) | = 2.3e-13 (i.e. exact).Overplotted RADIS-stack tau(nu,z) on the paper's aspect output for z = 0, 1, 2.5, 8 km.

Tools: RADIS 0.17, HITRAN, Python 3.11.

Scripts: `compare_gcell_vs_paper.py` , `compare_aspect_vs_radis_stack.py`

RESULTS - IS IT MATCHING? YES.

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gcell (gas cell) - RADIS vs paper's C code

Benchmark	paper tau_max	RADIS tau_max	rel-RMS*
02 A-band (Fig.4)	2.058	2.094	2.35 %
CH4 2.3 um (Fig.5)	6.409	6.395	7.65 %

aspect (atmosphere, US1976) - RADIS slab-stack vs paper's C code

partial column	paper tau_max	RADIS tau_max	rel-RMS*
TOA -> 0.0 km	570.27	563.61	2.67 %
TOA -> 1.0 km	542.04	534.96	2.75 %
TOA -> 2.5 km	499.17	491.40	3.18 %
TOA -> 8.0 km	339.58	329.94	4.45 %

* rel-RMS = root-mean-square relative difference, evaluated on the significant lines
(tau > 1% of the band maximum).

Interpretation

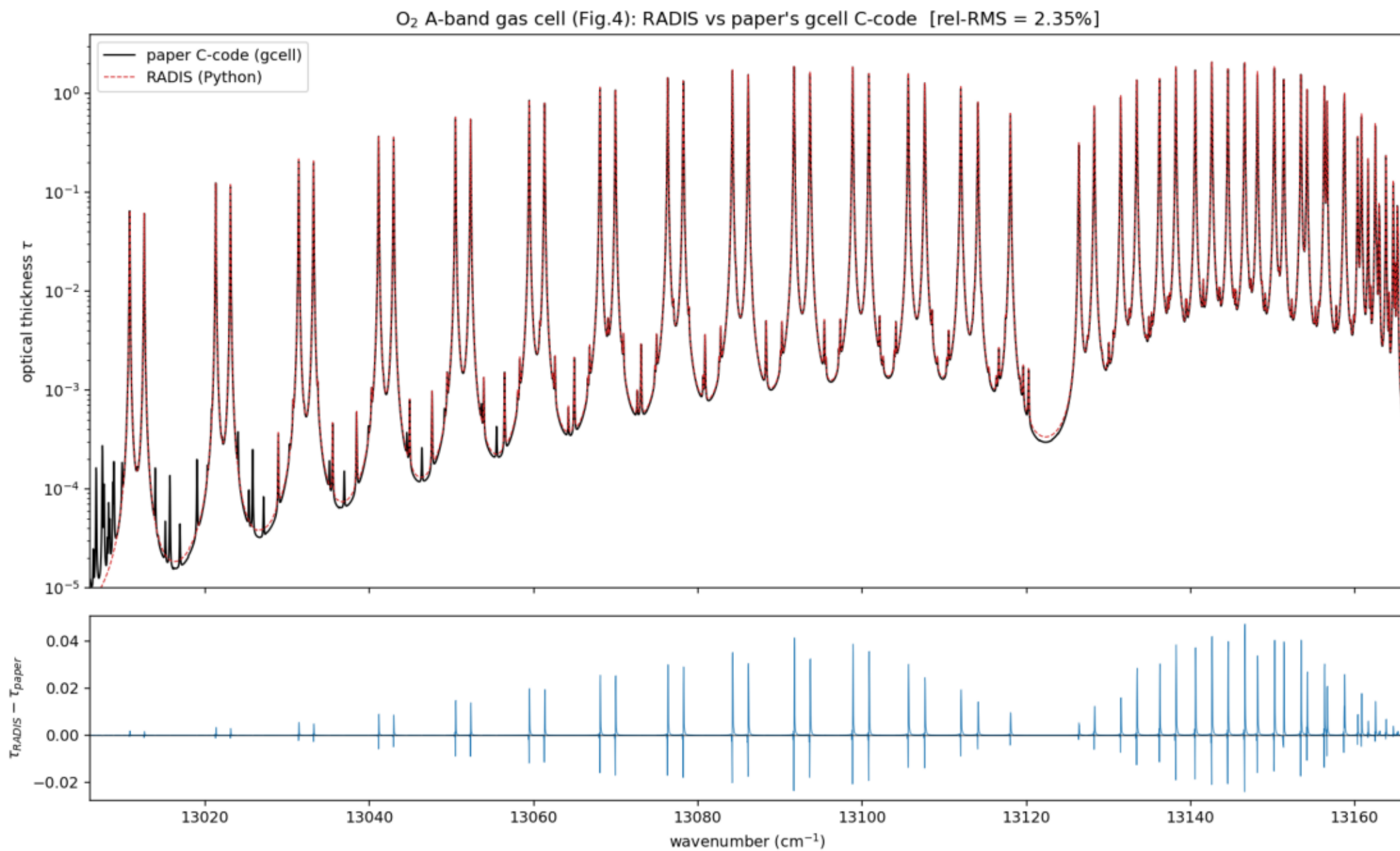
RADIS reproduces both codes line-by-line: line POSITIONS, DEPTHS, the band head and the height dependence all coincide (see the overplots on the following pages). The few-percent residual is NOT a modelling disagreement; it is consistent with:

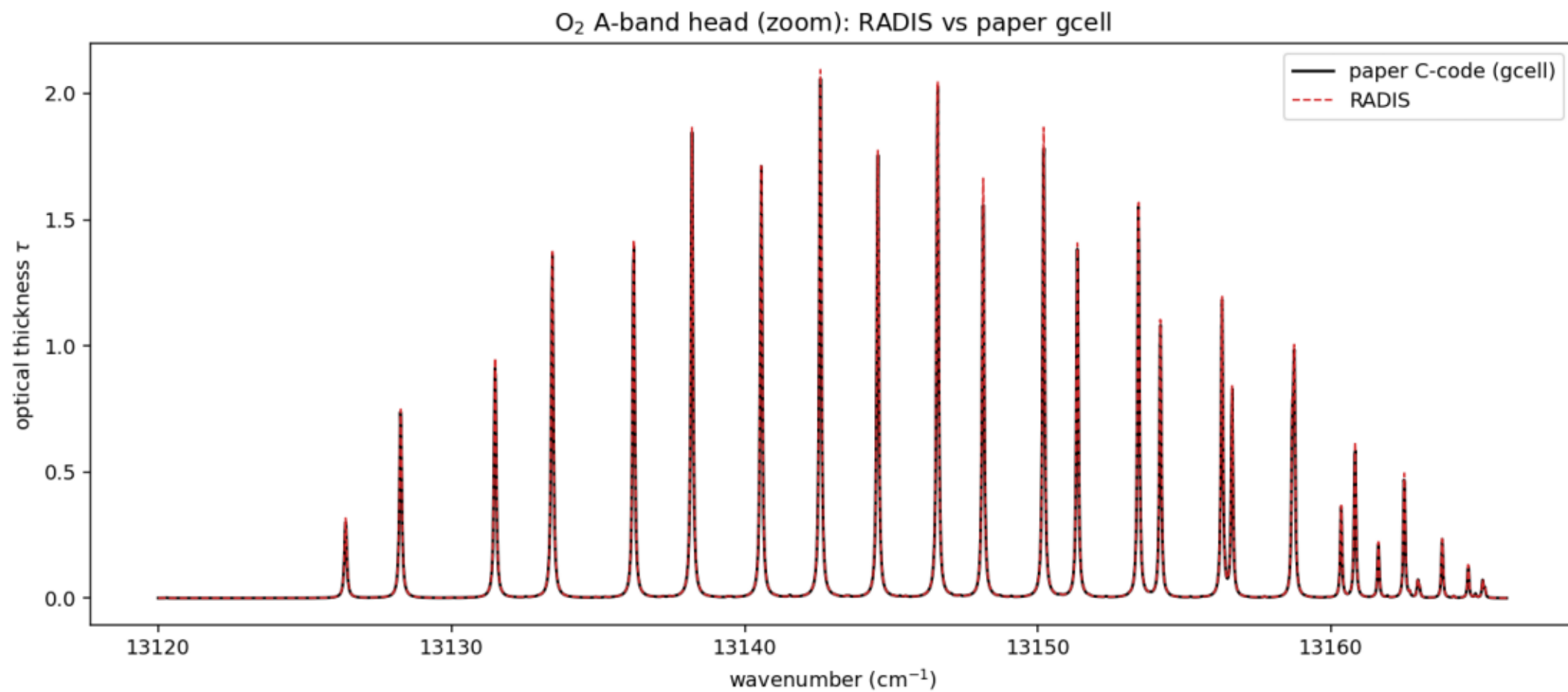
- (a) HITRAN version - the paper used HITRAN 2020, RADIS fetched the current HITRAN;
- (b) line-wing truncation - aspect cuts line wings at +/- 25 cm-1, RADIS uses its own default;
- (c) the slab discretisation of the continuous profile in the stack model.

The verified identity $\text{SerialSlabs} == \text{sum of optical depths}$ confirms the stack is the correct RADIS analogue of aspect's height integration.

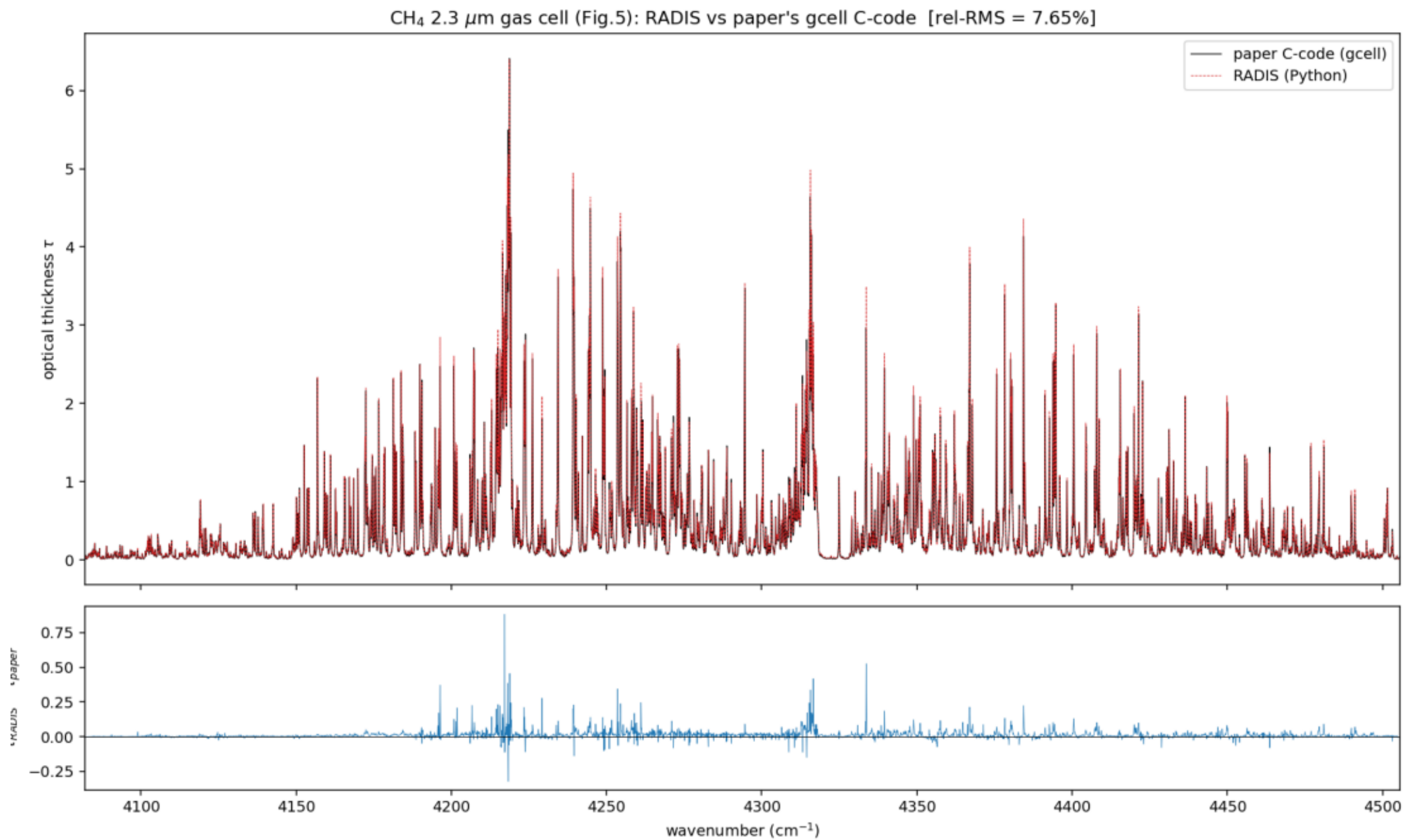
CONCLUSION: RADIS independently reproduces the paper's gcell and aspect results to within a few percent, validating both the paper's C codes and the RADIS workflow.

**gcell / Fig.4 - O₂ A-band gas cell: RADIS (red dashed) overplotted on the paper's C code (black).
Top: optical thickness (log scale). Bottom: residual. rel-RMS = 2.35%.**

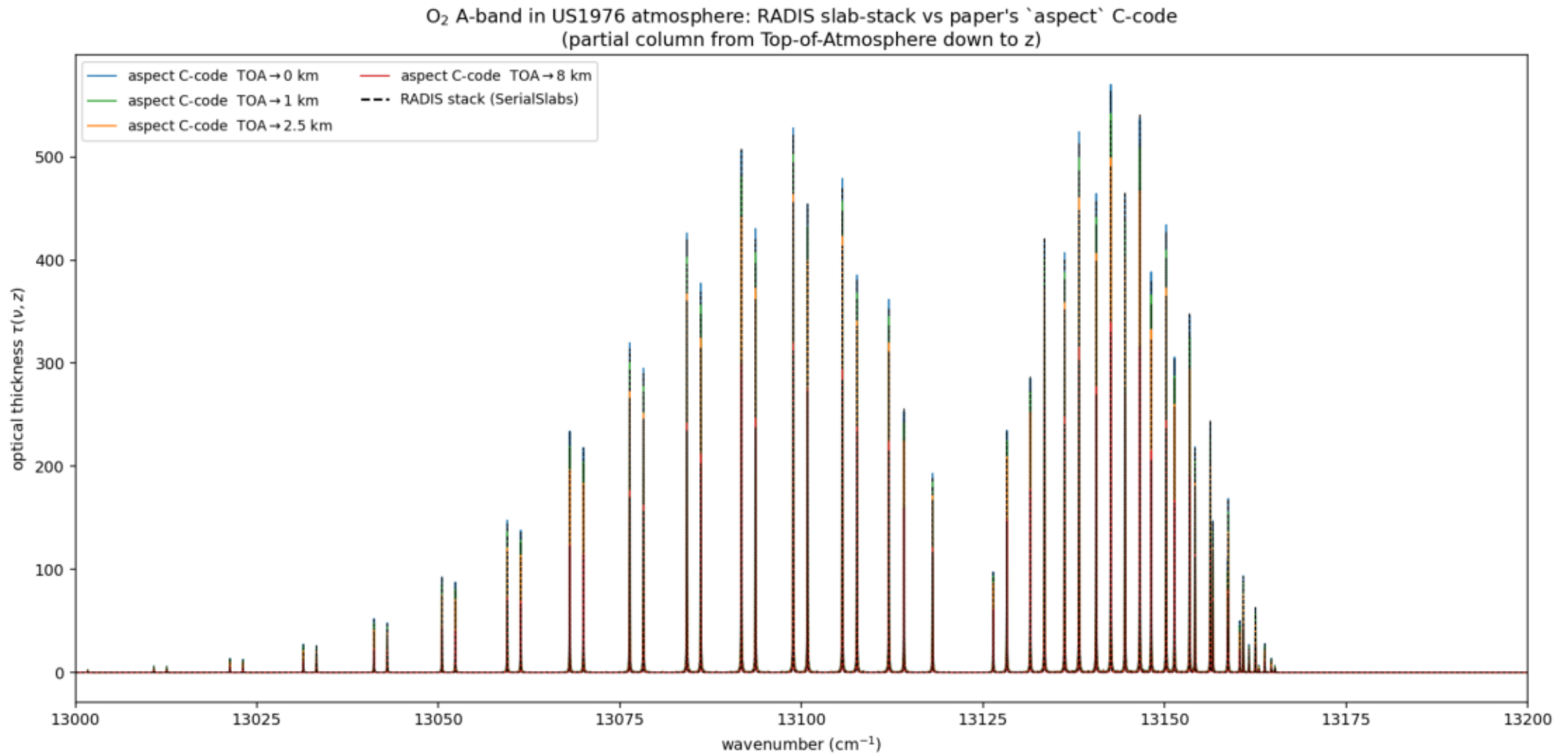




gcell / Fig.5 - CH₄ 2.3 μ m band: RADIS (red dashed) overplotted on the paper's C code (black).
Bottom: residual. rel-RMS = 7.65%.

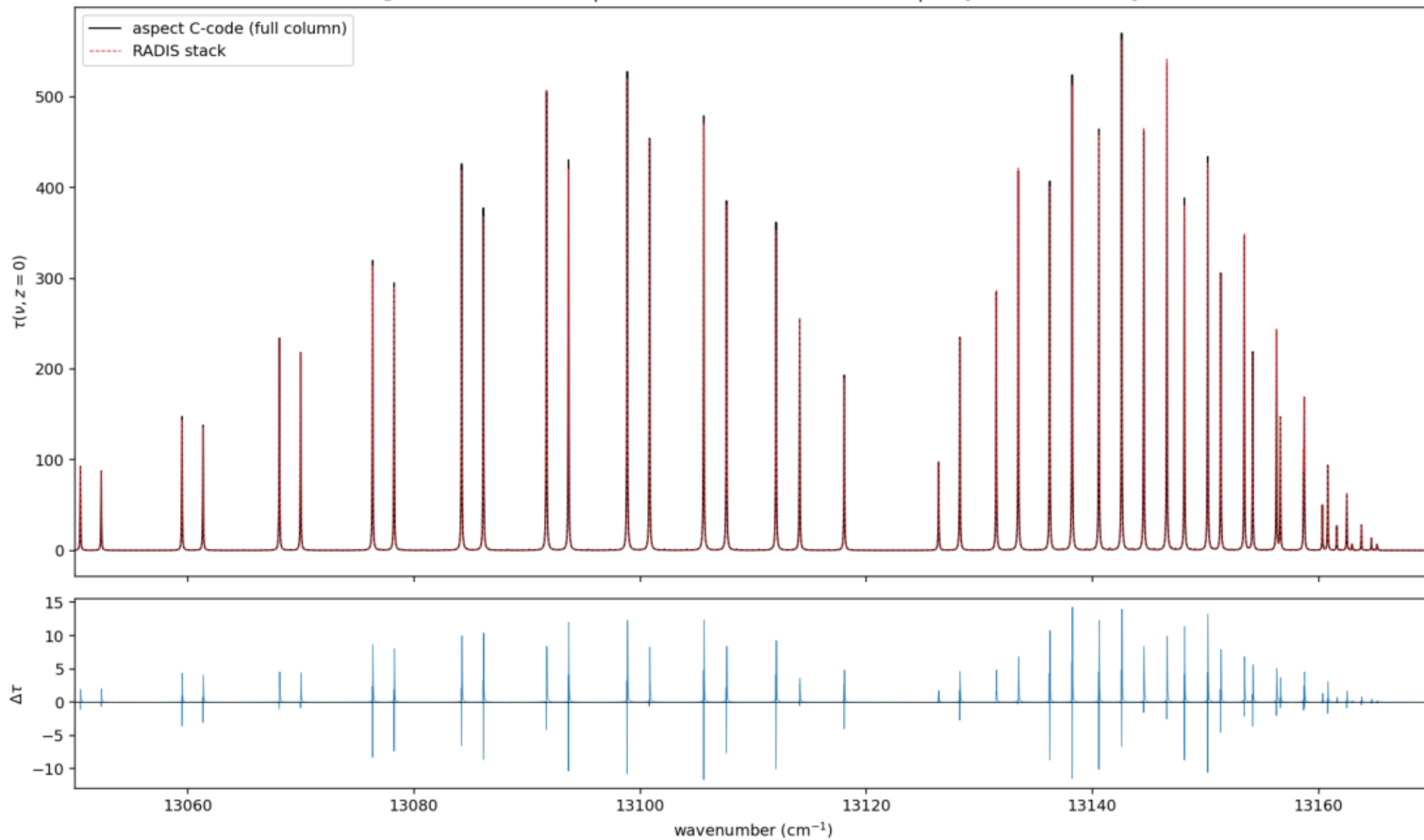


aspect - O₂ A-band in the US1976 atmosphere: RADIS slab-stack (black dashed) overplotted on the paper's aspect code (colours), for partial columns from TOA down to z = 0, 1, 2.5, 8 km.



aspect - Full-column (TOA->0 km) optical thickness, zoom + residual: RADIS stack vs aspect. rel-RMS = 2.67%.

O₂ A-band full-column optical thickness: RADIS stack vs aspect [rel-RMS = 2.67%]



aspect - Height variation: band-integrated O₂ A-band optical thickness vs altitude z.
RADIS stack (red) tracks the paper's aspect code (black) at every level.

